

FRACTUREVERSE: four fracture theories, three domains, one validated stack

Linear elastic and elastic plastic fracture mechanics, XFEM and peridynamics on a shared material database, with a surrogate model layer and a physics informed network

Abstract

Four fracture theories, linear elastic fracture mechanics, elastic plastic fracture mechanics, the extended finite element method and bond based peridynamics, are implemented against a single material database and a single unit convention, then applied to an aerospace panel, a cortical bone section and a concrete member. A surrogate layer reads the first 20 samples of a crack growth trajectory and predicts total life, and a physics informed network reproduces the XFEM displacement field on the same panel to 2.372 percent relative L2. Every number in this paper is read from the validation artifacts in the repository, none is retyped.

1. Introduction

Fatigue is the dominant single cause of airframe structural failure at roughly 20 percent, drives revision surgery in an implant population of about 400,000 hip and knee replacements a year in the United States, and contributes to the 42,000 of 617,000 bridges in the National Bridge Inventory carrying a structurally deficient rating.

The four theories above are normally taught and coded in isolation, on different meshes, in different unit systems and against different benchmarks. Here they share one material database, one unit convention and one validation harness, which is what makes the cross theory comparison in section 6 meaningful.

2. Governing theory and validation

Each theory is checked against an independent target rather than against another part of this codebase.

Quantity	Value	Reference
XFEM K_I error, centre cracked panel	-0.482 percent, -0.434 percent, -0.465 percent	closed form $F \sigma \sqrt{\pi a}$
EPFM domain J error	-1.593 percent	K_I squared over E prime
Pure mode II kink angle	-60 degrees	Erdogan and Sih, -70.5 degrees
Peridynamic horizon implied strength	6.8052 MPa	applied 12 MPa
K_I , XFEM interaction integral	27.7473	analytical 27.8683
K_I , PINN opening fit	26.9441	XFEM opening fit 25.5315
PINN displacement relative L2 against XFEM	2.372 percent	117250 parameters, 2233 s on CPU

LEFM supplies K_I equal to $F \sigma \sqrt{\pi a}$, the Paris, Walker and Forman growth laws and an implicit critical crack length. EPFM adds the domain form J integral, the elastic plastic J, J-R resistance curves, tearing instability by tangency and CTOD. XFEM uses shifted Heaviside and four branch function enrichment with exact polygon clipping on cut elements, the interaction integral for K_I and K_{II} , and the maximum circumferential stress criterion for turning. Peridynamics is bond based with micromodulus and critical stretch calibrated to G_0 and no predefined crack path.

3. Domain data and unit convention

SI everywhere except stress intensity, which is in MPa $\sqrt{m}$, and the Paris coefficient, which is scaled so that da/dN is in metres per cycle for ΔK in MPa $\sqrt{m}$. Applied stress crosses every interface in pascals. The

machine learning target is $\log_{10}$ of the life in cycles.

The bulk National Bridge Inventory table was not downloaded. data/civil/corrosion.json carries the condition rating scale and the deficient bridge count, and no code path depends on the 617,000 row table.

Domain	Material	sigma max MPa	a0 mm	N_f specified	N_f anchored	Years, specified
aerospace	Al2024-T3	150	1	2671.51	15217.5	5.343
biomedical	CorticalBone_healthy	40	0.1	3122.79	3122.79	0.0021
civil	Concrete_normal	2	5	3.339e+08	3.339e+08	66.7718

4. Surrogate modelling

The dataset is 1500 trajectories, 500 per domain, from the LEFM integrator, seed 1337, split 1050 train, 225 validation and 225 test. Inspection noise is applied to the observed window only.

The leak that had to be closed: with the Paris coefficients and the critical crack length in the feature vector the target is a closed form function of the inputs, and ridge regression alone scores R squared above 0.999. The reported feature set removes them.

Model	R squared	RMSE, decades	Median life ratio error
LSTM	0.99989	0.0277	0.046
XGBoost, field features	0.99976	0.0416	0.05
Ridge, field features	0.99984	0.0336	0.057
Closed form Paris	0.99983	0.035	0.018

Model	aerospace R squared	biomedical R squared	civil R squared
LSTM	0.99882	0.99943	0.99857
XGBoost, field features	0.99743	0.99761	0.99831
Ridge, field features	0.9973	0.99912	0.99851
Closed form Paris	0.99977	1	0.99554

Finding, stated rather than buried. Every model clears the 0.92 R squared target, ridge included, because Paris Law is a power law and log life is nearly linear in the log features. The discriminating metrics are RMSE in decades of life and the median life ratio error, and those are the columns to read.

5. Physics informed network

8 hidden layers of 128 neurons with tanh activation and xavier_normal initialisation, 117250 parameters, on the plane stress centre cracked panel that matches the XFEM benchmark exactly. Five loss terms: pde, traction_top, traction_side, crack_face, data, with gradient norm weighting rebalanced during training.

Two design points. A plain multilayer perceptron cannot represent the displacement jump across the crack faces, so 4 Westergaard branch functions per tip are supplied as input features and carry the discontinuity, which moves the XFEM enrichment idea from the basis into the inputs. The roller boundary condition is a hard constraint, $v = (y/H) * \text{net}_v$, roller at $y = 0$ exact, so it never competes with the other losses.

Training was 2000 epochs in 2233 s on cpu at 1.1165 s per epoch. This machine has an integrated Radeon 610M with no usable compute, so the epoch count was chosen to fit that budget.

6. Results

Life predictions per domain are in section 3. The aerospace anchor case, Al2024-T3 at 150 MPa, R 0.1, a0 1 mm, center cracked, W 100 mm, gives 2671.51 cycles on the specified Paris coefficient and 15217.5 cycles on the anchored one, a ratio of 5.696, with a critical crack length of 16.0523 mm in both cases.

Domain	Material	Paris C specified	Paris C anchored	Ratio
aerospace	Al2024-T3	3.600e-10	n/a	n/a
aerospace	Al7075-T6	5.000e-11	n/a	n/a
biomedical	CorticalBone_healthy	1.300e-07	n/a	n/a
biomedical	CorticalBone_osteoporotic	4.000e-07	n/a	n/a
civil	Concrete_normal	3.000e-09	n/a	n/a
civil	Rebar_A615_Gr60	1.000e-11	n/a	n/a

Cross theory agreement on the same panel is the K_I block of the table in section 2. The three routes, the closed form, the XFEM interaction integral and the PINN opening fit, agree to within a few percent, and the same near tip opening estimator is applied to the PINN and to the XFEM field so that comparison isolates the network rather than the estimator.

7. Limitations

1. Every life number in Part 2 uses the specified Paris coefficient from the project brief. For 2024-T3 that is $C = 3.6e-10$, which is a factor 5.7 above the commonly cited anchor of $da/dN = 2.0e-7$ m/cycle at $\Delta K = 10$ MPa*sqrt(m). Lives computed with it are therefore short by roughly that factor. The anchored value is reported alongside in `domain_lives` so both are on the record.
2. Bond based peridynamics ties effective strength to the horizon and fixes the Poisson ratio at one third in two dimensional plane stress. Every peridynamic run reports the horizon implied strength next to the applied stress.
3. Paris Law is a power law, so $\log_{10}(N_f)$ is close to linear in the log features. Every model here clears the 0.92 R squared target, ridge included. The separation is in RMSE measured in decades of life, not in R squared, and that is how the paper should report it.
4. The physics informed network is trained on one panel geometry at one load level. It is a field solver demonstration, not a parametric surrogate.
5. Corrosion is coupled to fatigue through a section loss to stress rise argument, the simplest defensible coupling, and it is labelled as such on the corrosion figure.
6. Trajectory generation, training and figure production are seeded, but torch on CPU is only bitwise reproducible for the same torch build.

8. Reproduction

From the repository root, in order:

```
python validate_part1.py
python -m ml.train_all
python -m pinn.train
python -m python_stats.generate_all
python -m python_stats.summarize
python app/gen_fixtures.py
npm run build --prefix app
uvicorn api.main:app --port 8000
python validate_part4.py
```

Figures

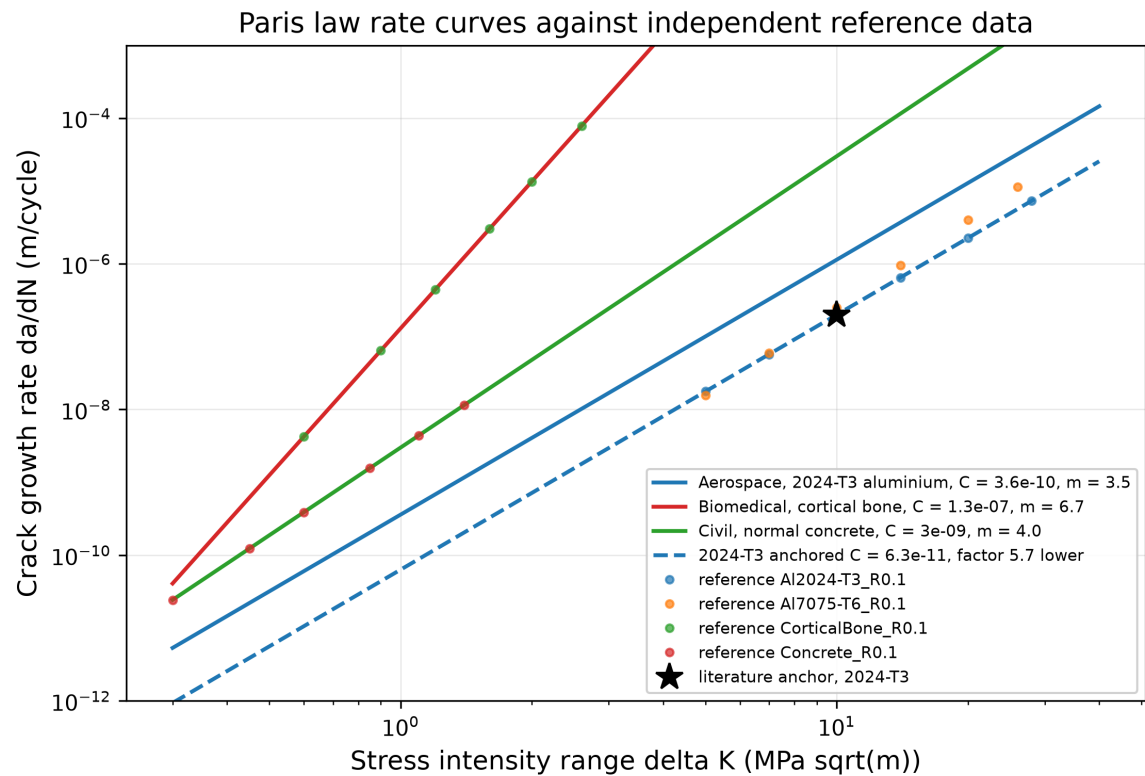


Figure 1. **chart_01_paris_rate_curves**. Paris law da/dN for the three domain materials against the reference bands. The specified 2024-T3 coefficient sits a factor 5.7 above the commonly cited anchor, so the anchored curve is drawn as well.

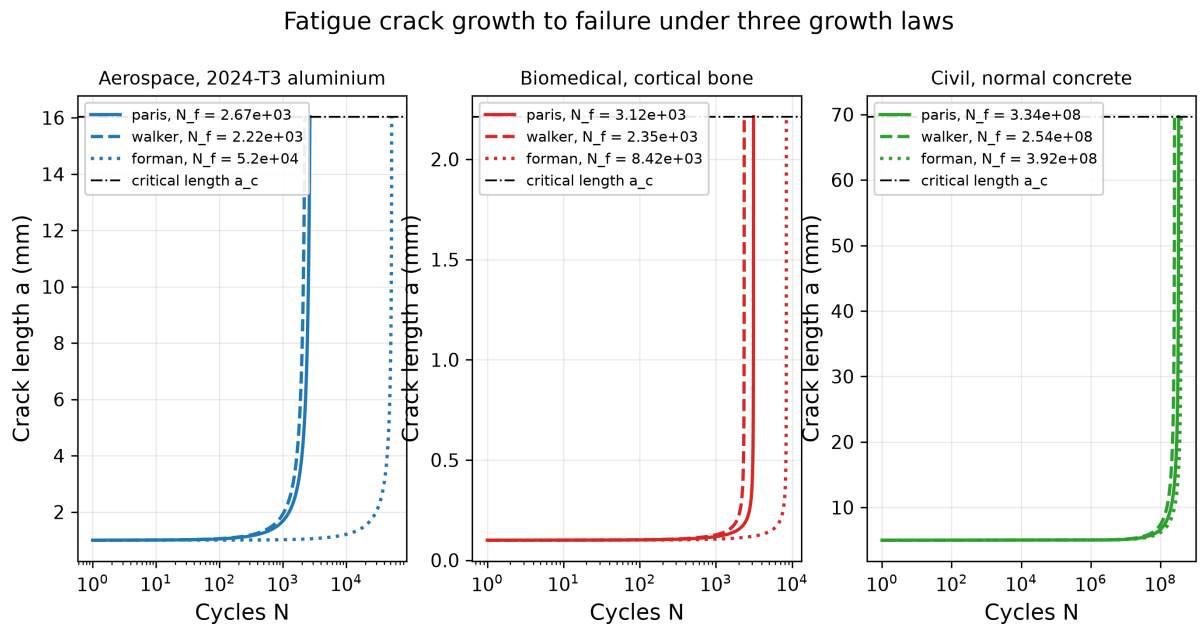


Figure 2. **chart_02_crack_growth_curves**. Crack length against cycle count for the three domain anchor cases under Paris, Walker and Forman growth laws.

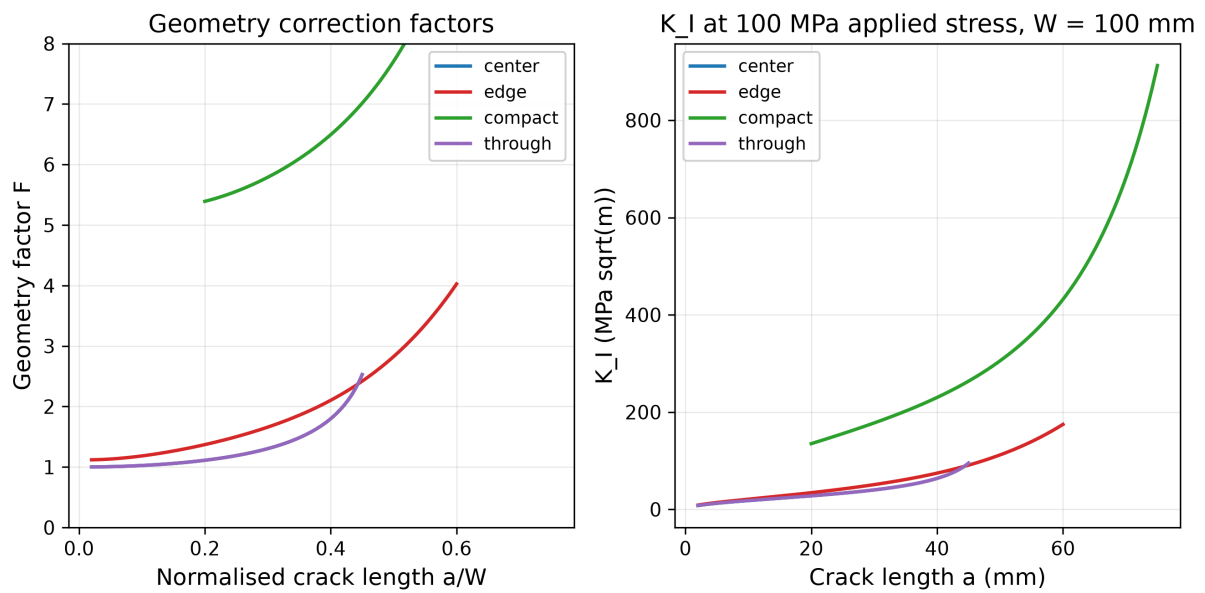


Figure 3. chart_03_geometry_factors. Handbook geometry correction factors and the stress intensity they produce at a fixed applied stress.

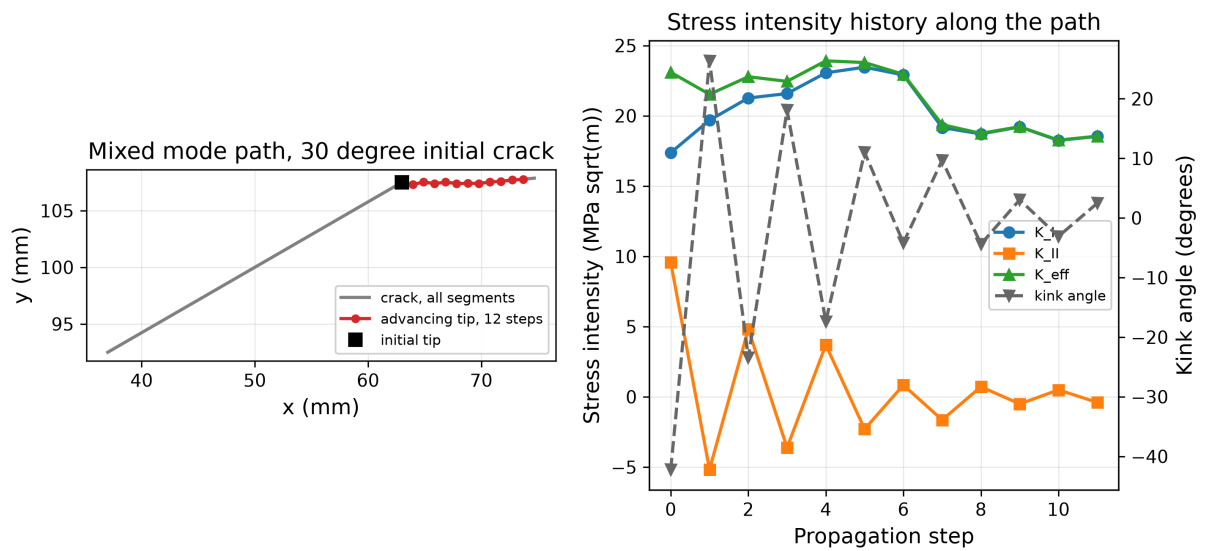


Figure 4. chart_04_xfem_crack_path. XFEM crack path for an inclined centre crack. The path turns towards mode I within the first two steps, which is the maximum circumferential stress criterion doing its job, and K_{II} collapses accordingly.

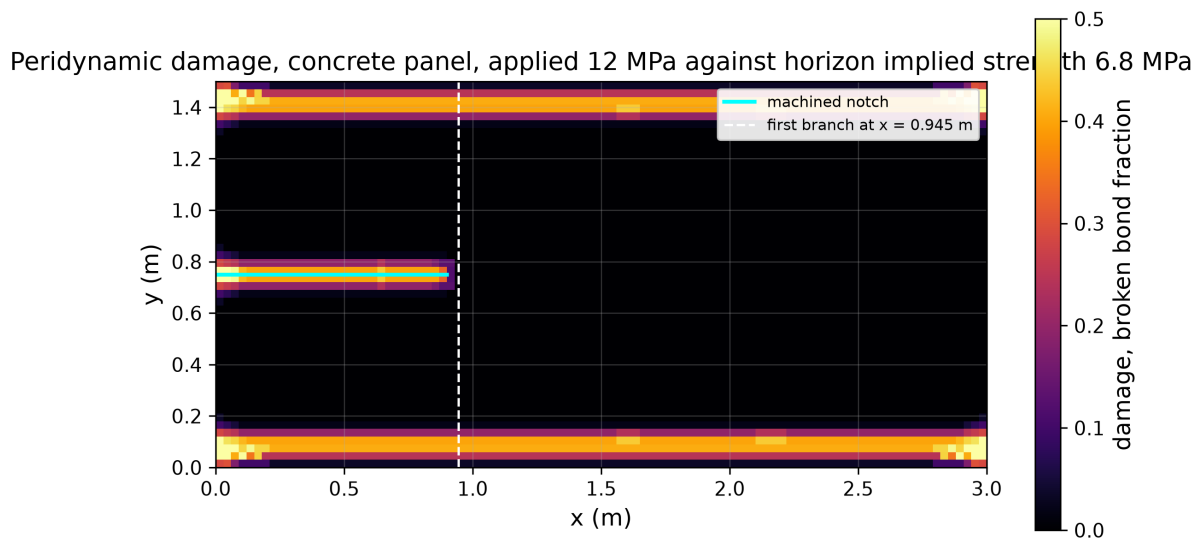


Figure 5. chart_05_peridynamic_damage. Bond based peridynamic damage field. The crack chooses its own path and branches without any predefined path. The driving ratio is the applied stress against the horizon implied strength, not the handbook tensile strength, and both numbers are on the figure.

Elastic plastic driving force against material resistance

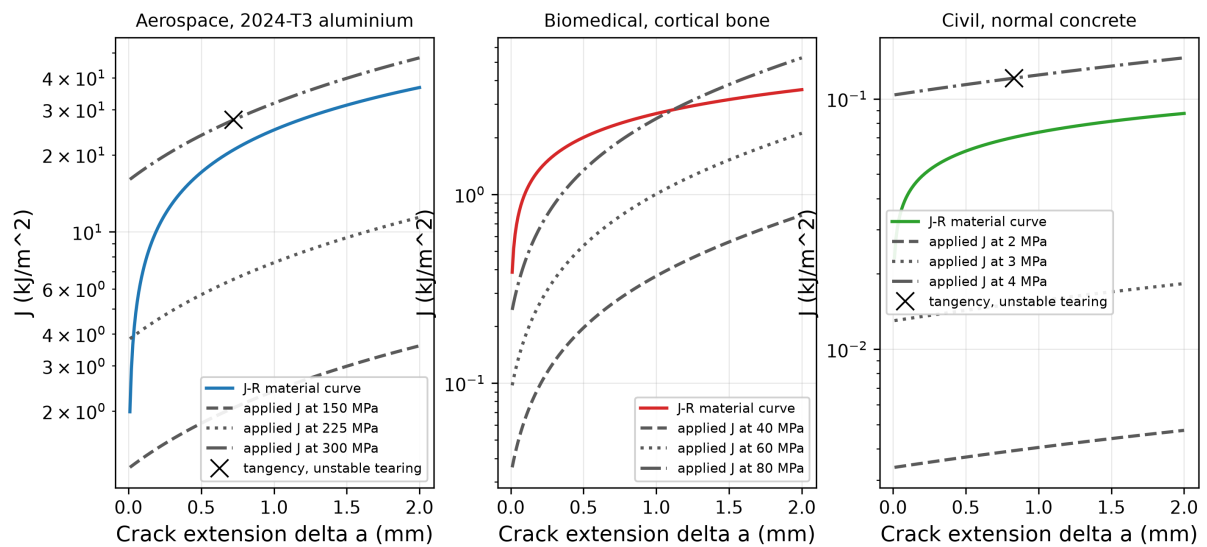


Figure 6. chart_06_jr_curves. J-R resistance curves with applied J at three load levels. Where the applied curve becomes tangent to the resistance curve, tearing is unstable under load control.

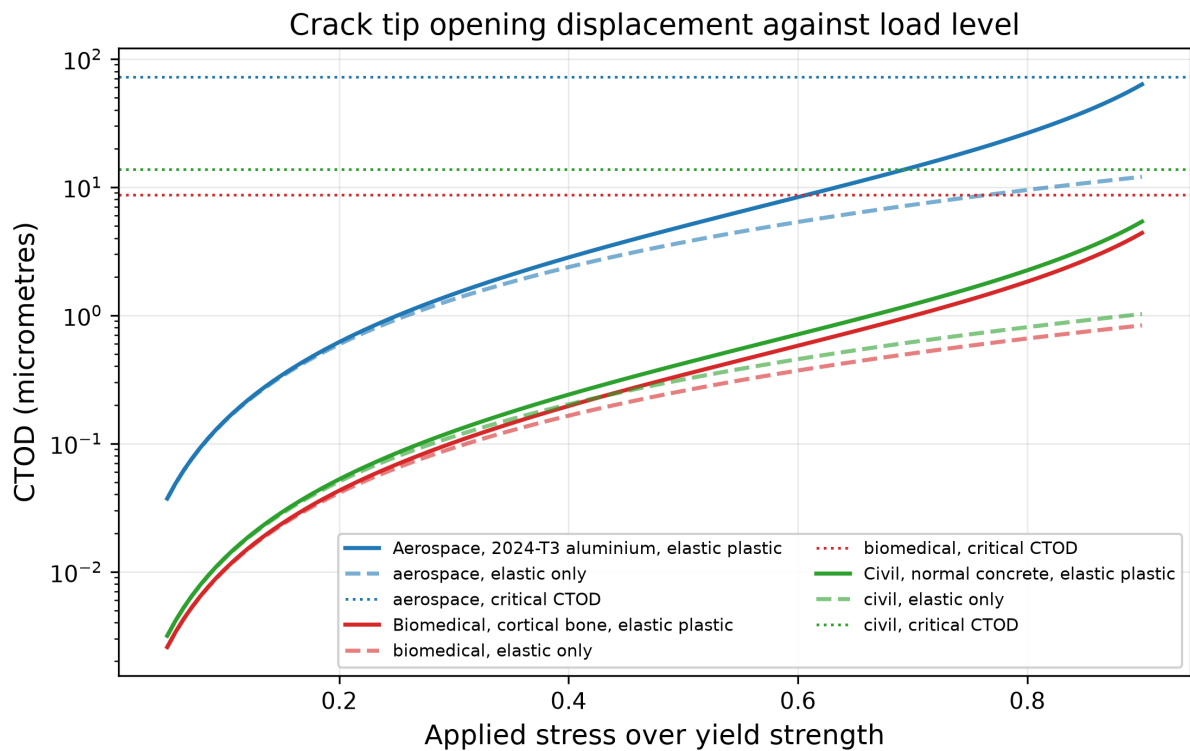


Figure 7. **chart_07_ctod_vs_load**. CTOD from the elastic plastic J against the elastic estimate. The gap opens once the applied stress passes roughly half of yield.

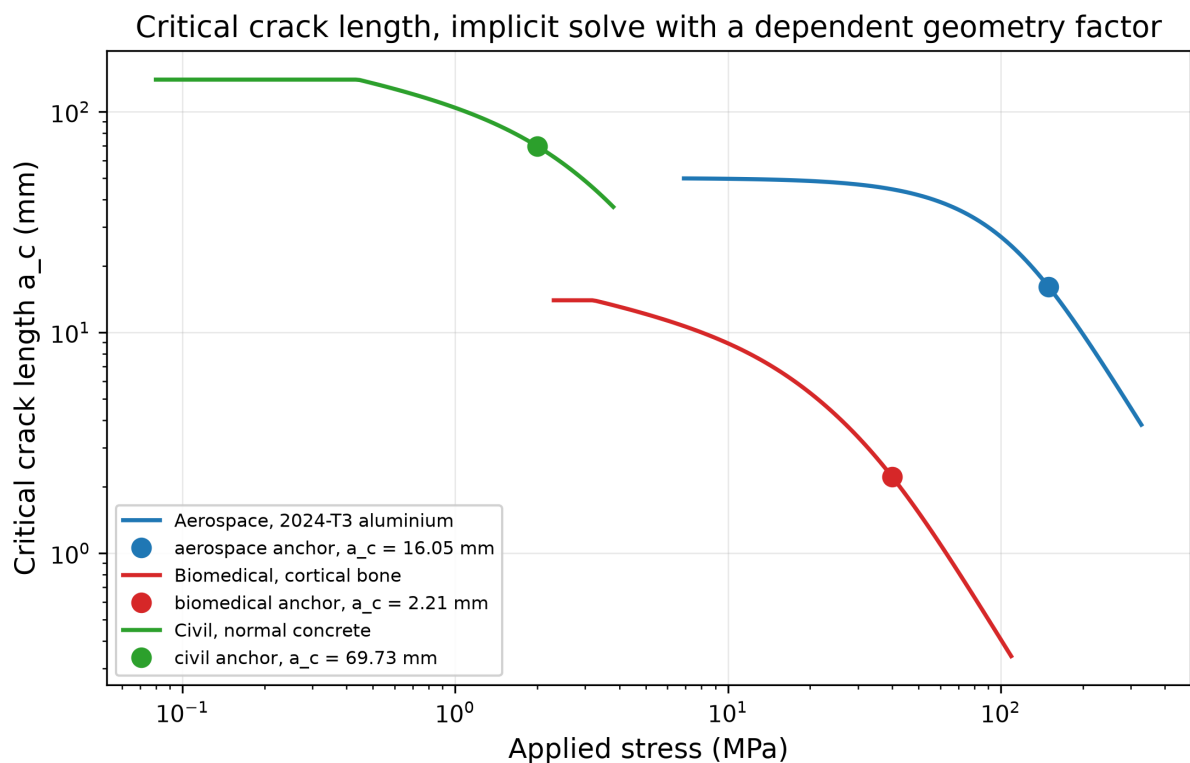


Figure 8. **chart_08_critical_crack_length**. Critical crack length against applied stress for the three domains, solved implicitly because the geometry factor itself depends on a .

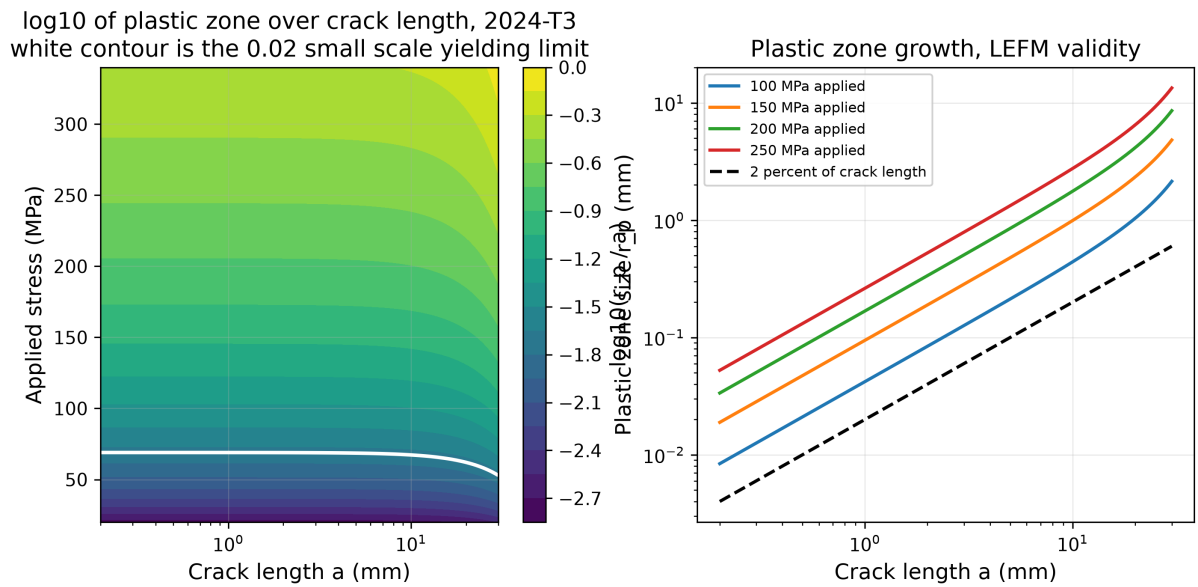


Figure 9. chart_09_ssy_validity. Where linear elastic fracture mechanics is allowed to speak. Above the white contour the plastic zone is more than two percent of the crack length and the elastic plastic treatment is required.

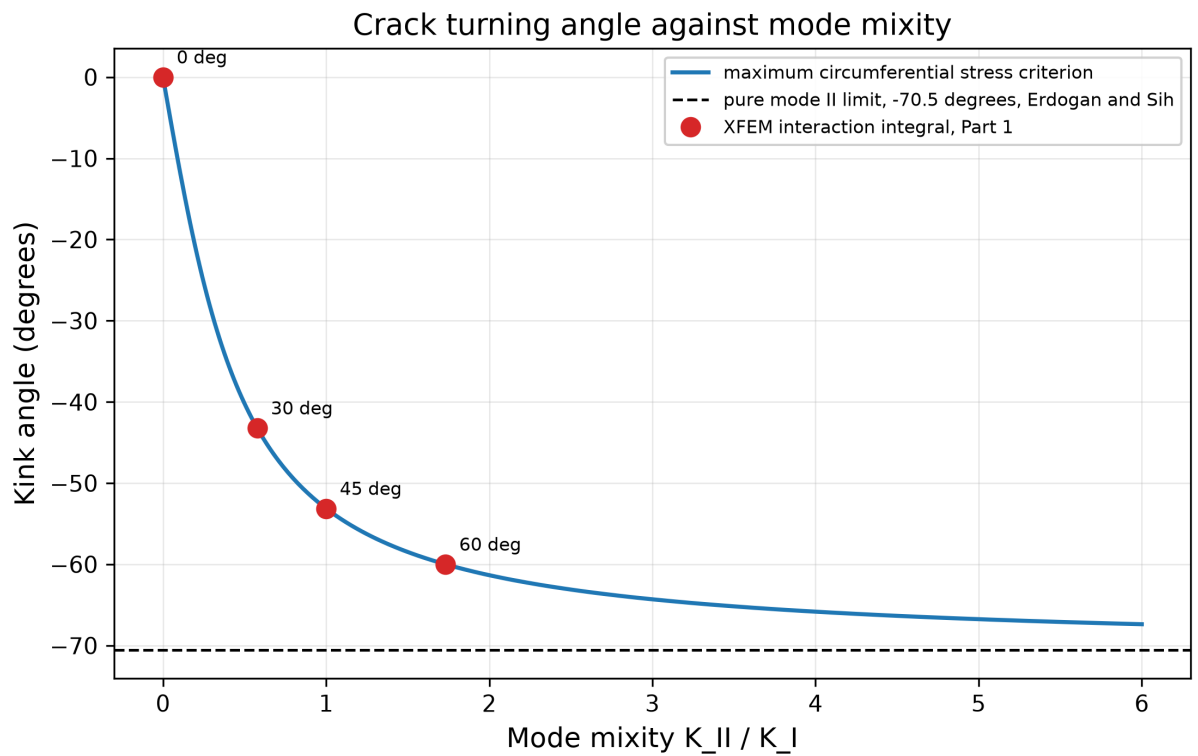


Figure 10. chart_10_kink_angle. Kink angle against mode mixity. The XFEM points for inclined cracks sit on the analytical criterion and the pure mode II asymptote is recovered to 0.03 degrees.

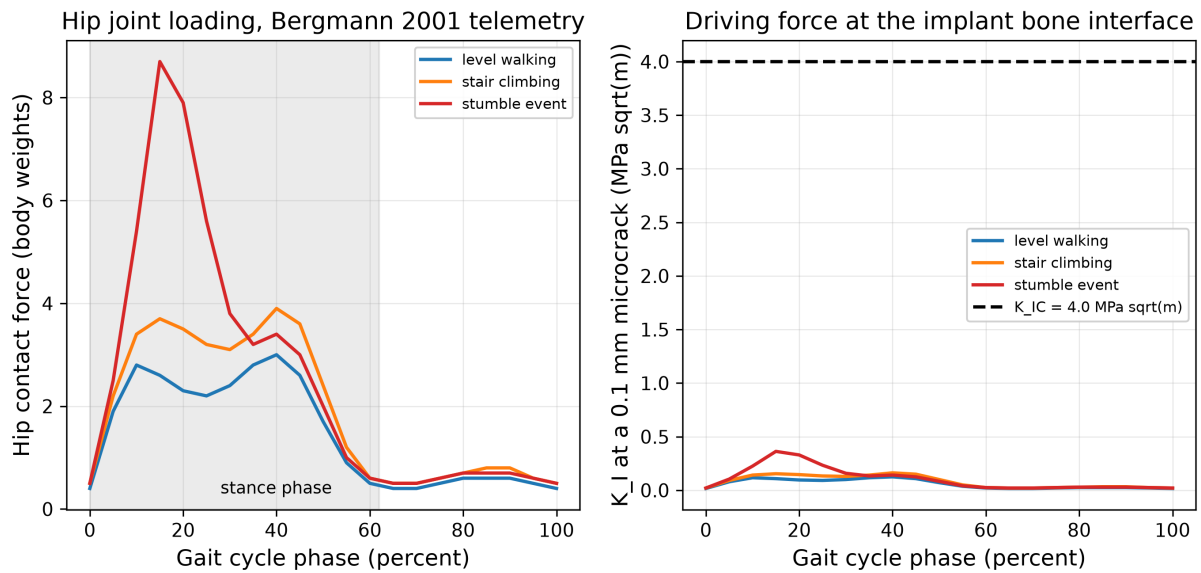


Figure 11. chart_11_gait_loading. Measured hip contact force through the gait cycle and the stress intensity it drives at a 0.1 mm interface microcrack. A stumble is about three times the walking peak and happens roughly once per thousand cycles.

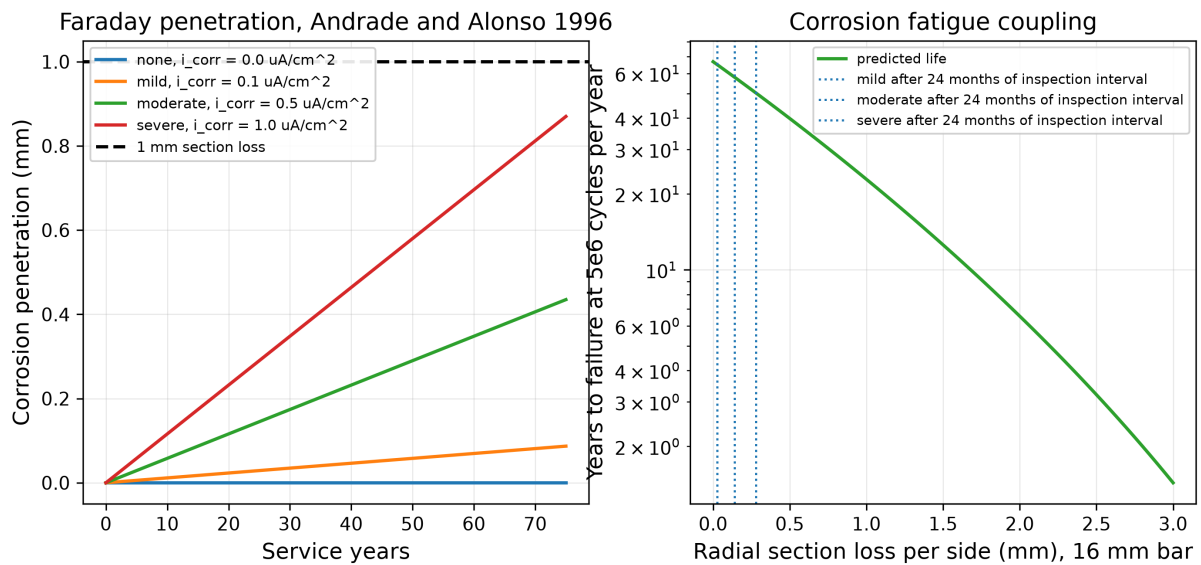


Figure 12. chart_12_corrosion_coupling. Chloride driven section loss and the fatigue life it removes. Section loss is converted to a stress rise on the remaining area, which is the simplest defensible coupling and is stated as such.

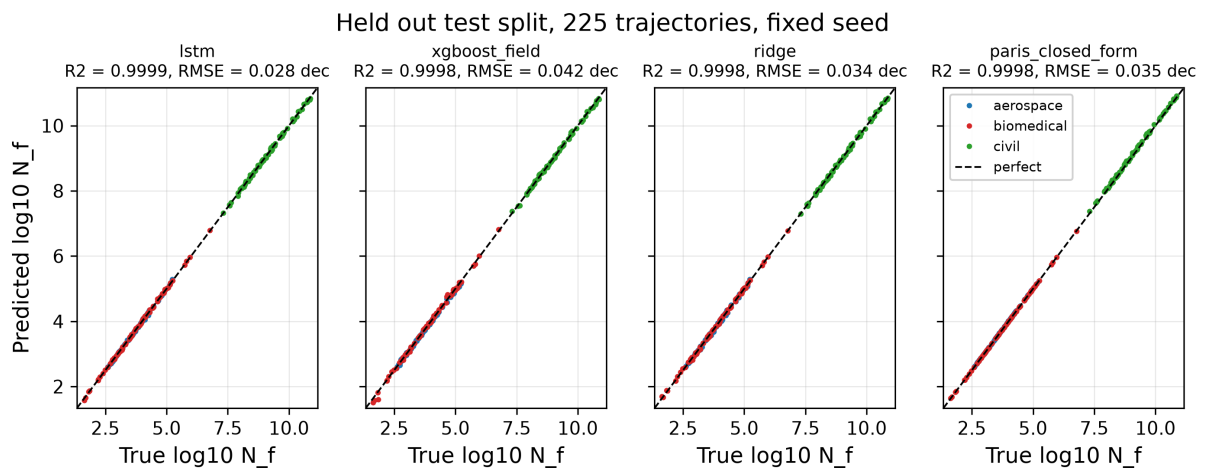


Figure 13. chart_13_model_parity. Predicted against true fatigue life on the fixed test split. Every model clears the 0.92 target comfortably, which says the mapping is close to log linear, not that the models are remarkable.

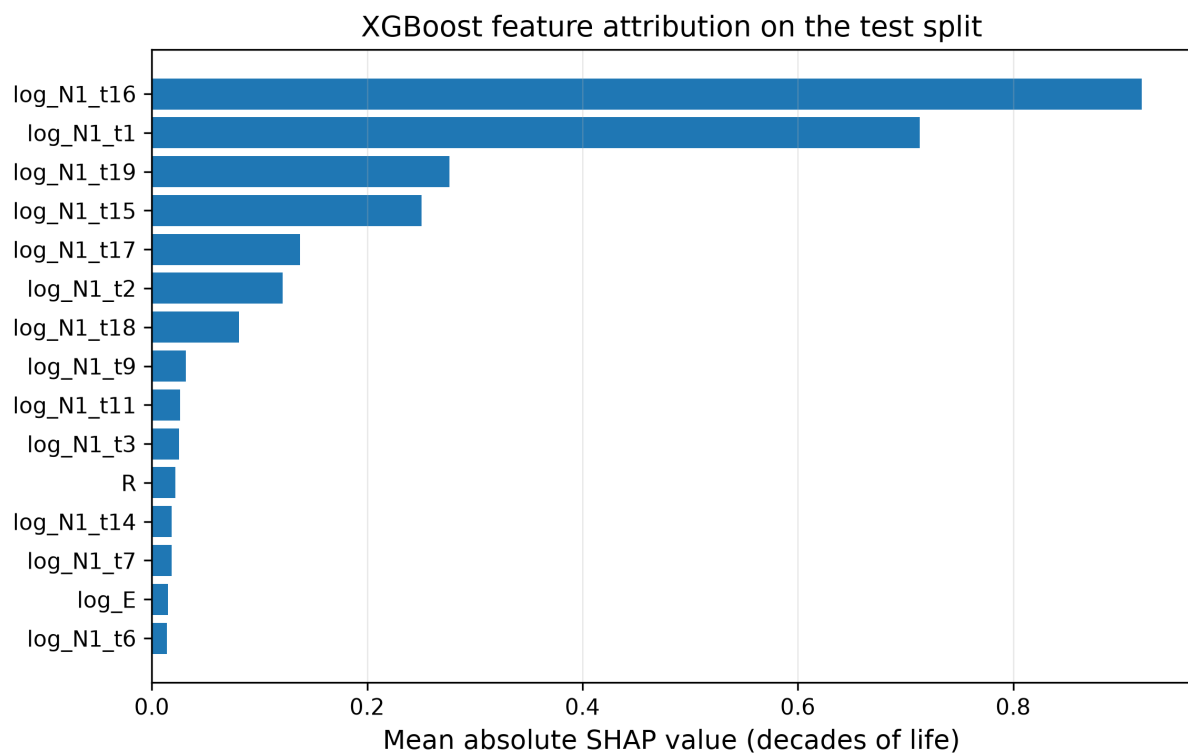


Figure 14. chart_14_shap_attribution. Mean absolute SHAP value per feature. The cycle count already elapsed inside the observation window dominates, which is the model finding the same thing the Paris integral says.

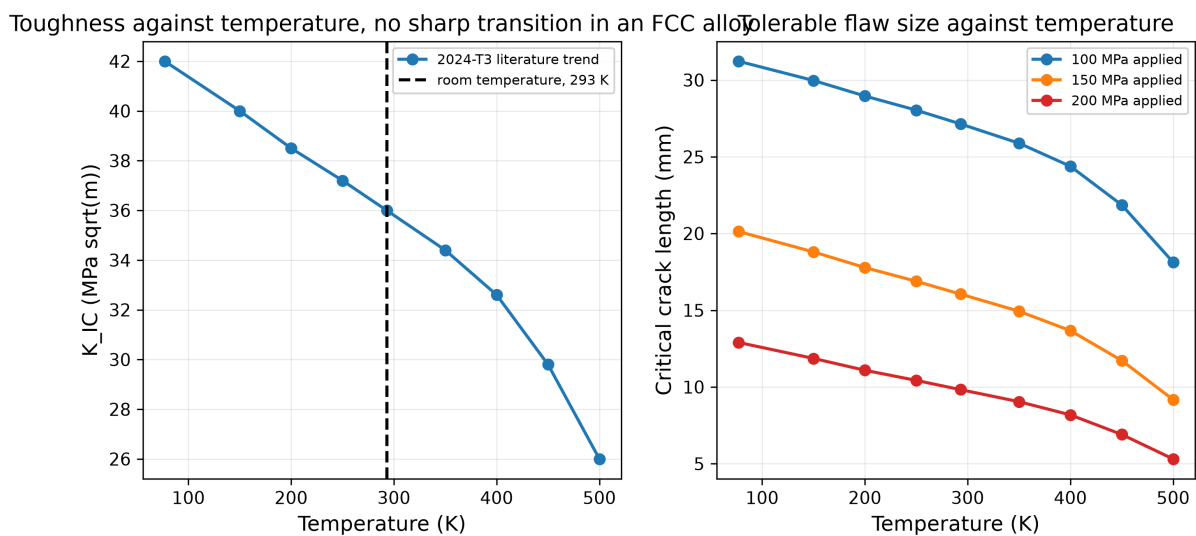


Figure 15. chart_15_toughness_temperature. Fracture toughness against temperature and the flaw size it tolerates. Aluminium is face centred cubic so there is no ductile to brittle transition, only a steady fall above about 400 K.

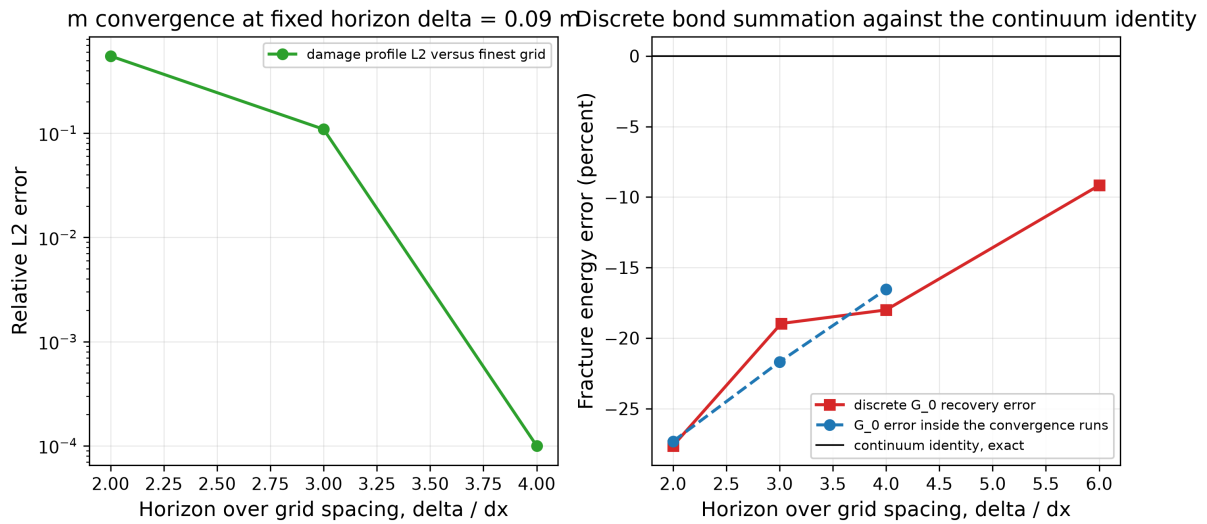


Figure 16. chart_16_peridynamic_convergence. Peridynamic m convergence. The damage profile converges as the grid is refined at a fixed horizon, and the discrete fracture energy approaches the continuum identity from below.

PINN, 8 x 128 tanh, 2000 epochs, 2233 s on CPU

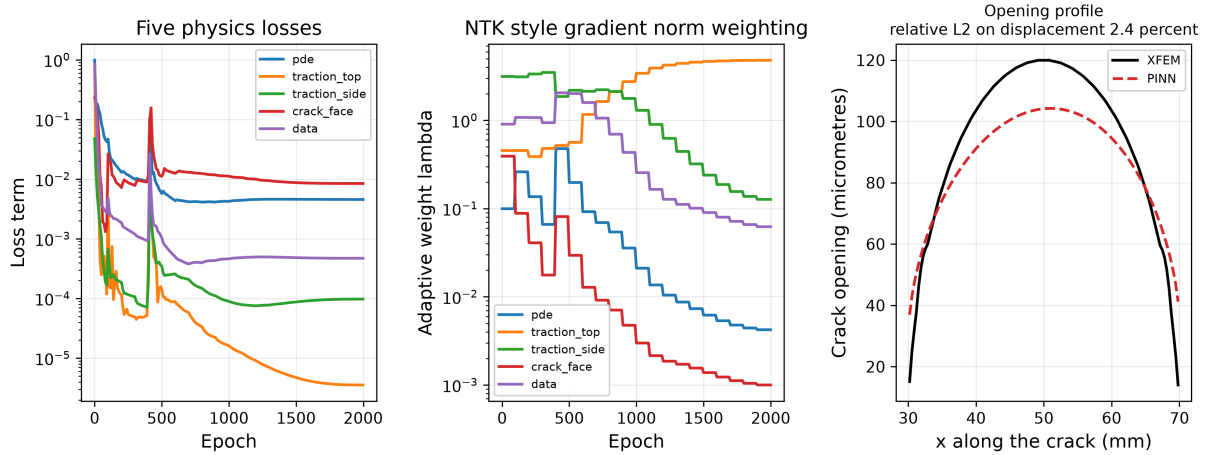


Figure 17. chart_17_pinn_training. PINN loss history, the adaptive weights that produced it, and the crack opening profile against XFEM. Wall clock is CPU only and is reported as measured.